ICU Critical Care

Final Exam Study Guide

High Yield Notes + Practice MCQs

ملاحظات مركزة + أسئلة تدريبية اختيارية

PART 1: Cardiac & Respiratory Emergencies

1. Acute Coronary Syndrome (ACS)

ACS is a spectrum of conditions caused by reduced blood flow to the heart muscle, ranging from unstable angina to complete blockage (STEMI).

مجموعة حالات ناتجة عن نقص التروية في القلب، من الذبحة غير المستقرة حتى الاحتشاء الكامل.

Types of ACS

Stable Angina	Chest pain with exertion → relieved by rest or nitroglycerin. No plaque rupture.
Unstable Angina	Plaque rupture → chest pain at rest. More dangerous than stable angina.
NSTEMI	Partial coronary occlusion → troponin elevated, no ST elevation on ECG.
STEMI	Complete occlusion by clot → ST elevation on ECG. Requires urgent PCI or thrombolysis.

☆ Key Clinical Points

- **Chest pain radiating to the shoulder → FIRST do: ECG (رسم قلب)**
- **Before hospital: Aspirin 300 mg (4 tablets) + Clopidogrel (Plavix) 4 tablets**
- PCI = Percutaneous Coronary Intervention (cardiac catheterization to open artery)
- **STEMI Treatment: PCI OR Thrombolytic Therapy**
- ⚠ **NEVER give Nitroglycerin if patient took Viagra → severe hypotension!**

ACS Symptoms

- Chest Pain — Shortness of Breath — Palpitations

- Sweating — Nausea — Vomiting

Diagnosis Tools

- ECG — Troponin — CK-MB — Echocardiography

2. Stroke

A stroke occurs when blood supply to part of the brain is cut off (ischemic) or a blood vessel bursts (hemorrhagic).

جلطة الدماغ: إما انسداد وعاء (إقفاري) أو نزيف (نزفي).

Types of Stroke

Ischemic Stroke	Thrombotic or Embolic. Most common type.
Hemorrhagic Stroke	Intracerebral Hemorrhage OR Subarachnoid Hemorrhage

☆ Stroke Management — Step by Step

- **Patient with arm/leg weakness (suspected stroke) → FIRST step: Non-Contrast CT Brain**
- **Why CT first? To differentiate Ischemic vs Hemorrhagic — treatment differs!**
- ❌ **Do NOT give Aspirin or Heparin before CT result**
- Ischemic stroke may NOT appear on CT in first 24–48 hours
- If CT normal + still suspicious → repeat CT after 48 hours OR do MRI with Diffusion
- **MRI with Diffusion = best test for EARLY detection of ischemic stroke**

Blood Pressure Targets in Stroke

Ischemic Stroke (first 24–48 h)	Maintain BP < 180/105 mmHg (do NOT lower aggressively — preserve penumbra)
Hemorrhagic Stroke	Target Systolic BP < 140 mmHg

Penumbra = الأنسجة القابلة للإنقاذ حول الجلطة — لذلك نحافظ على الضغط مرتفعاً نسبياً في الجلطة الإقفارية

Stroke Risk Factors

- Hypertension — Diabetes — Dyslipidemia
- Atrial Fibrillation (AF) — Carotid Stenosis ⭐

Stroke Complications

- **Aspiration Pneumonia — most common cause of DEATH in stroke patients** ⭐
- DVT — Fever — Septicemia

3. Cardiac Arrhythmias

Bradycardia	Heart Rate < 60 bpm. Causes: Athletes, sleep, Beta Blockers
Tachycardia	Heart Rate > 100 bpm. Causes: Fever, Shock, Pain, Anxiety, Exercise
Atrial Flutter	Presence of F-waves (sawtooth) on ECG
Atrial Fibrillation (AF)	No P-waves on ECG. Biggest risk: Ischemic Stroke ⭐
Ventricular Fibrillation	Immediately life-threatening → TREATMENT: DC Shock ⭐⭐
Pulseless VT	Ventricular Tachycardia without a pulse → TREATMENT: DC Shock

SVT Vagal Maneuvers (to slow heart)

- Carotid Sinus Massage
- Valsalva Maneuver
- Cold Water on Face
- Breath Holding
- Pressure on Eyes

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4. Respiratory Failure & COPD

SpO_2 Normal	> 95%
COPD Target SpO_2	88–92% ⭐ Do NOT exceed this!

☆ Why NOT give high O_2 in COPD?

COPD patients rely on HYPOXIA as their breathing drive. If you raise SpO_2 to 99%, they may stop breathing → Apnea → Respiratory Arrest!

مرضى COPD يعتمدون على نقص الأكسجين كمحفز للتنفس. رفع الأكسجين كثيراً يوقف التنفس!

CO_2 Monitoring	ABG (Arterial Blood Gas) OR Capnogram device

PART 2: Neurological & ICU Monitoring

5. Intracranial Pressure (ICP) & CPP

Normal ICP	< 15 mmHg

Target CPP	> 70 mmHg

Formula	CPP = MAP − ICP

How to Raise CPP?

- Lower the ICP — OR — Raise MAP (blood pressure)

Nursing Care to REDUCE ICP

- Elevate Head of Bed: 30–45°
- Keep neck straight (Straight Neck) — improves venous drainage from brain
- Maintain CPP > 70 mmHg
- Keep all Vital Signs normal
- Seizure Prophylaxis

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6. Glasgow Coma Scale (GCS)

Full Score	15 points (Eye 4 + Verbal 5 + Motor 6)

Score	Eye (E) /4	Verbal (V) /5	Motor (M) /6
4	Spontaneous	Oriented	Obeys Commands
3	To Voice	Confused	Localizes Pain
2	To Pain	Words	Withdraws
1	None	Sounds	Abnormal Flexion
—	—	None (1)	Abnormal Extension
—	—	—	None (1)

☆ **GCS < 8 → Perform Endotracheal Intubation (patient cannot protect airway)**

GCS أقل من 8 = يحتاج تنبيب لأنه لا يستطيع حماية مجرى الهواء

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7. Hemodynamic Monitoring — Normal Values

Parameter	Normal Range	Notes
MAP (Mean Arterial Pressure)	> 65 mmHg	Minimum acceptable
Heart Rate (Pulse)	60–100 bpm	Tachy>100, Brady<60
Temperature	36.5–37.2 °C	Fever in ICU > 38.5°C ⭐
Respiratory Rate	12–20 / min	
SpO_2 (Oxygen Saturation)	> 95%	COPD: 88–92% ⭐
Urine Output	0.5–1 ml/kg/hr	70kg patient = 35–70 ml/hr
Fasting Blood Sugar	< 120–125 mg/dL	
Post-meal Blood Sugar (2h)	< 180 mg/dL	

☆ First test for any ICU admission: Random Blood Sugar (RBS)

PART 3: VAP Bundle, Shock & DKA

8. VAP Bundle (Ventilator-Associated Pneumonia Prevention)

The VAP Bundle is a set of 5 interventions that MUST ALL be applied together to prevent pneumonia in ventilated patients.

مجموعة 5 إجراءات لازم تتطبق كلها مع بعض لمنع الالتهاب الرئوي في المريض على جهاز التنفس.

#	Intervention	Details
1	Elevation of Head of Bed	Keep at > 45° (reduces aspiration risk) ⭐
2	Oral Care	Regular mouth hygiene to reduce bacterial load
3	Daily Sedation Vacation	Stop sedation daily to assess readiness for extubation
4	DVT Prophylaxis	Heparin / Enoxaparin (Clexane) / Pneumatic Compression Pump
5	Stress Ulcer Prophylaxis	Use PPI (Proton Pump Inhibitor)

9. Shock — Types & Management

Hypovolemic	Cardiogenic	Obstructive	Distributive
Hemorrhage Gastroenteritis Dehydration	Heart pump failure May cause Bradycardia	PE Pneumothorax Cardiac Tamponade ⭐	Septic Neurogenic (Warm!) Anaphylactic

- ⭐ **ALL shock types → Tachycardia EXCEPT: Cardiogenic Shock & Neurogenic Shock (may cause Bradycardia)**
- **Distributive Shock → patient is WARM (due to vasodilation + increased capillary permeability)**

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10. DKA — Diabetic Ketoacidosis

Diagnostic Triad ☆

Hyperglycemia	Random Blood Sugar > 250 mg/dL
Ketosis	Ketones present in blood/urine
Acidosis	pH < 7.3 AND HCO_3 < 15

DKA Management

- Insulin dose: 0.1 unit/kg/hour IV infusion (e.g. 80 kg patient → 8 units/hr)
- Target glucose drop: 50–80 mg/dL per hour (gradual!)
- Most important electrolyte to monitor: POTASSIUM ⭐
- ⭐ **Most dangerous complication of DKA: HYPOGLYCEMIA (dropping sugar too fast)**

Classic Diabetes Symptoms

- Polyuria (frequent urination)
- Polydipsia (excessive thirst)
- Weight Loss

🩺 PART 4: Hepatic Encephalopathy, Hypertension & GI Bleeding

11. Hepatic Encephalopathy

Brain dysfunction caused by liver failure and accumulation of ammonia in the blood.

اضطراب في وظائف المخ ناتج عن فشل الكبد وتراكم الأمونيا في الدم.

Precipitating Factors (causes of episodes)

- Infection — Constipation — Dehydration
- High Protein Intake (↑ ammonia production)
- Diuretics — GI Bleeding

Stages of Consciousness Deterioration

- Stage 1: Poor Concentration
- Stage 2: Delirium
- Stage 3: Drowsiness
- Stage 4: Coma

Treatment ☆

- **Decrease Protein Intake**
- **Lactulose — prevents ammonia absorption ⭐ ⭐**
- **Gastrobiotic (Rifaximin)**
- **Enema (bowel cleaning) ⭐ — Doctor emphasized this strongly**
- Treat underlying infection

DVT Prevention in Hepatic Patients

- **Use: Pneumatic Compression Pump ⭐**
- **❌ AVOID anticoagulants — these patients have low platelets + high INR**

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12. Hypertensive Crisis

Hypertensive Emergency	BP > 180/120 + TARGET ORGAN DAMAGE → lower BP GRADUALLY

Hypertensive Urgency	BP > 180/120, NO organ damage (e.g. Epistaxis = nosebleed) ⭐

Eclampsia (special)	BP > 160/105 in pregnant women → Emergency ⭐

Target Organ Damage includes:

- Hypertensive Encephalopathy — Stroke — Pulmonary Edema
- ACS — Acute Renal Failure — Aortic Dissection — Eclampsia

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13. Upper GI Bleeding

Hematemesis	Melena	Hematochezia	Occult Blood
Vomiting Blood قيء دم	Black Tarry Stool براز أسود	Fresh Blood Per Rectum دم أحمر من الشرج	Hidden Blood in Stool دم خفي

Common Causes of GI Bleeding

- Peptic Ulcer — Erosions — Mallory-Weiss Syndrome — Tumors

Drugs that Cause GI Bleeding

- Aspirin — NSAIDs — Heparin — Warfarin — Clopidogrel

PART 5: Thyroid Emergencies

14. Thyroid Disorders

	Hypothyroidism (↓ function)	Hyperthyroidism (↑ function)
TSH	↑ (high)	↓ (low)
Weight	↑ Weight gain	↓ Weight loss
Metabolism	Everything slows ↓	Everything speeds ↑
Heart Rate	Slow / Brady	Fast / Tachy
Temperature	Cold intolerance	Heat intolerance

QUICK REFERENCE — Most Tested Values

ICP Normal	< 15 mmHg
CPP Target	> 70 mmHg
CPP Formula	CPP = MAP − ICP
MAP Minimum	> 65 mmHg
SpO_2 Normal	> 95%
COPD SpO_2 Target	88–92%
ICU Fever Definition	> 38.5 °C
Urine Output	0.5–1 ml/kg/hr

GCS Full Score	15 (E4+V5+M6)
GCS → Intubation	GCS < 8
VAP HOB Angle	> 45°
ICP Nursing HOB	30–45°
DKA Sugar Cutoff	> 250 mg/dL
DKA pH	< 7.3
DKA HCO_3	< 15 mEq/L
HTN Emergency	BP > 180/120 + Organ Damage
HTN Urgency	BP > 180/120, NO organ damage
Eclampsia	BP > 160/105
STEMI Treatment	PCI or Thrombolysis
VF Treatment	DC Shock ⭐ ⭐

📝 FINAL EXAM — MCQ Practice Questions (50 Questions)

Answer all questions. The correct answer is marked in green after each question.

اجب على كل الأسئلة. الإجابة الصحيحة مكتوبة باللون الأخضر تحت كل سؤال.

1. A patient presents with chest pain radiating to the left shoulder. What is the FIRST thing to do?

A) Give Aspirin immediately
B) Perform ECG
C) Give Nitroglycerin
D) Start oxygen

Answer: B) Perform ECG

2. What is the normal range for Intracranial Pressure (ICP)?

A) > 20 mmHg
B) < 15 mmHg
C) 15–25 mmHg
D) < 10 mmHg

Answer: B) < 15 mmHg

3. What is the CPP formula?

A) CPP = MAP + ICP
B) CPP = HR × MAP
C) CPP = MAP − ICP
D) CPP = ICP − MAP

Answer: C) CPP = MAP − ICP

4. What is the target CPP in ICU patients?

A) > 50 mmHg
B) > 60 mmHg
C) > 70 mmHg
D) > 80 mmHg

Answer: C) > 70 mmHg

5. The VAP Bundle head elevation angle is:

A) 15–20°
B) 30°
C) 30–45°
D) > 45°

Answer: D) > 45°

6. Which of these is NOT part of the VAP Bundle?

A) Oral Care
B) Daily Sedation Vacation
C) Stress Ulcer Prophylaxis
D) Chest Physiotherapy

Answer: D) Chest Physiotherapy

7. What is Daily Sedation Vacation?

A) Giving sedation every day
B) Stopping sedation daily to assess readiness to wean ventilator
C) Changing the sedation drug daily
D) Giving sedation only at night

Answer: B) Stopping sedation daily to assess readiness to wean ventilator

8. Which drug is used for DVT Prophylaxis?

A) Aspirin
B) Metoprolol
C) Enoxaparin (Clexane)
D) Furosemide

Answer: C) Enoxaparin (Clexane)

9. How do we prevent Stress Ulcers in ICU?

A) H2 blockers only
B) PPI (Proton Pump Inhibitor)
C) Antibiotics
D) Steroids

Answer: B) PPI (Proton Pump Inhibitor)

10. The full GCS score is:

A) 10
B) 12
C) 15
D) 20

Answer: C) 15

11. GCS Motor Response 'Localizes Pain' scores:

A) 6
B) 5
C) 4
D) 3

Answer: B) 5

12. At what GCS level should Endotracheal Intubation be performed?

A) GCS < 12
B) GCS < 10
C) GCS < 8
D) GCS < 6

Answer: C) GCS < 8

13. The minimum acceptable MAP in ICU is:

A) > 55 mmHg
B) > 60 mmHg
C) > 65 mmHg
D) > 75 mmHg

Answer: C) > 65 mmHg

14. The target SpO_2 in COPD patients is:

A) 95–99%
B) 90–94%
C) 88–92%
D) 85–89%

Answer: C) 88–92%

15. Why do we NOT give high oxygen to COPD patients?

A) It causes infection
B) Hypoxia is their breathing drive; high O_2 can cause apnea
C) They are allergic to oxygen
D) It raises blood pressure

Answer: B) Hypoxia is their breathing drive; high O_2 can cause apnea

16. Normal Urine Output in ICU is:

A) 0.1–0.3 ml/kg/hr
B) 0.5–1 ml/kg/hr

C) 1–2 ml/kg/hr
D) 2–3 ml/kg/hr
Answer: B) 0.5–1 ml/kg/hr

17. For a 70 kg patient, what is the normal hourly urine output?
A) 10–20 ml/hr
B) 20–35 ml/hr
C) 35–70 ml/hr
D) 70–140 ml/hr
Answer: C) 35–70 ml/hr

18. The first test to order when a STEMI is suspected:
A) Troponin
B) Echocardiography
C) ECG
D) Chest X-ray
Answer: C) ECG

19. Treatment for STEMI is:
A) Aspirin only
B) Heparin drip
C) PCI or Thrombolytic therapy
D) Beta blockers
Answer: C) PCI or Thrombolytic therapy

20. Which drug combination is DANGEROUS together due to severe hypotension?
A) Aspirin + Heparin
B) Nitroglycerin + Viagra
C) Clopidogrel + Aspirin
D) Metoprolol + Atenolol
Answer: B) Nitroglycerin + Viagra

21. A patient with suspected stroke presents with arm weakness. What is the FIRST step?
A) Give Aspirin
B) Give Heparin
C) Non-Contrast CT Brain
D) MRI Brain
Answer: C) Non-Contrast CT Brain

22. Why do we do a CT scan before any treatment in stroke?
A) It's faster than MRI
B) To differentiate ischemic from hemorrhagic stroke
C) To measure ICP
D) To check for DVT
Answer: B) To differentiate ischemic from hemorrhagic stroke

23. Which is the BEST early test to detect Ischemic Stroke?
A) Non-contrast CT
B) MRI with Diffusion
C) CT with contrast
D) Angiography
Answer: B) MRI with Diffusion

24. Target BP in Hemorrhagic Stroke:

A) SBP < 180
B) SBP < 160
C) SBP < 140
D) SBP < 120

Answer: C) SBP < 140

25. The most common cause of DEATH in stroke patients is:

A) DVT
B) Aspiration Pneumonia
C) Renal failure
D) Cardiac arrest

Answer: B) Aspiration Pneumonia

26. Atrial Fibrillation is characterized by:

A) F-waves (sawtooth)
B) No P-waves
C) Widened QRS
D) ST elevation

Answer: B) No P-waves

27. The most dangerous complication of Atrial Fibrillation is:

A) Ventricular Fibrillation
B) Ischemic Stroke
C) Heart Block
D) Pulmonary Embolism

Answer: B) Ischemic Stroke

28. Treatment for Ventricular Fibrillation is:

A) Atropine IV
B) Amiodarone
C) DC Shock
D) CPR alone

Answer: C) DC Shock

29. Which type of Shock has WARM extremities?

A) Hypovolemic Shock
B) Cardiogenic Shock
C) Obstructive Shock
D) Distributive Shock

Answer: D) Distributive Shock

30. Which Shock types can cause BRADYCARDIA instead of Tachycardia?

A) Hypovolemic and Septic
B) Cardiogenic and Neurogenic
C) Anaphylactic and Obstructive
D) Distributive and Hypovolemic

Answer: B) Cardiogenic and Neurogenic

31. What are the causes of Obstructive Shock?

A) Hemorrhage, dehydration, vomiting
B) PE, Pneumothorax, Cardiac Tamponade
C) Sepsis, anaphylaxis, neurogenic
D) Heart failure, arrhythmia

Answer: B) PE, Pneumothorax, Cardiac Tamponade

32. DKA Diagnostic Triad includes:

A) Hypoglycemia + Alkalosis + Ketonuria
B) Hyperglycemia + Ketosis + Acidosis
C) Hyperglycemia + Hypokalemia + Alkalosis
D) Hypoglycemia + Ketosis + Acidosis

Answer: B) Hyperglycemia + Ketosis + Acidosis

33. Blood sugar level for DKA diagnosis is:

A) > 100 mg/dL
B) > 150 mg/dL
C) > 200 mg/dL
D) > 250 mg/dL

Answer: D) > 250 mg/dL

34. Insulin dose in DKA:

A) 0.01 unit/kg/hr
B) 0.05 unit/kg/hr
C) 0.1 unit/kg/hr
D) 0.5 unit/kg/hr

Answer: C) 0.1 unit/kg/hr

35. Most dangerous electrolyte to monitor in DKA:

A) Sodium
B) Potassium
C) Calcium
D) Magnesium

Answer: B) Potassium

36. Most dangerous complication of DKA treatment:

A) Hypernatremia
B) Hypoglycemia
C) Alkalosis
D) Hyperkalemia

Answer: B) Hypoglycemia

37. What prevents Hepatic Encephalopathy most effectively?

A) High protein diet
B) Lactulose
C) Antibiotics only
D) Diuretics

Answer: B) Lactulose

38. The most emphasized nursing intervention in Hepatic Encephalopathy:

A) IV fluids
B) Enema (bowel cleaning)
C) Nasogastric feeding

D) Oxygen therapy

Answer: B) Enema (bowel cleaning)

39. Why does HIGH protein intake cause Hepatic Encephalopathy?

A) It damages the liver directly

B) It increases ammonia production

C) It causes hyponatremia

D) It reduces albumin

Answer: B) It increases ammonia production

40. DVT prevention in Hepatic patients should use:

A) Heparin injection

B) Warfarin

C) Pneumatic Compression Pump

D) Aspirin

Answer: C) Pneumatic Compression Pump

41. Hypertensive Emergency is defined as:

A) BP > 140/90 with symptoms

B) BP > 160/100 without organ damage

C) BP > 180/120 with target organ damage

D) BP > 200/100 always

Answer: C) BP > 180/120 with target organ damage

42. Epistaxis (nosebleed) with BP 185/115 is classified as:

A) Hypertensive Emergency

B) Hypertensive Urgency

C) Normal variant

D) Hypertensive Crisis requiring IV drugs

Answer: B) Hypertensive Urgency

43. Eclampsia is defined as Hypertensive Emergency when:

A) BP > 140/90

B) BP > 160/105

C) BP > 180/120

D) BP > 200/110

Answer: B) BP > 160/105

44. Hematemesis means:

A) Blood in urine

B) Vomiting blood

C) Black stools

D) Blood in sputum

Answer: B) Vomiting blood

45. Melena is:

A) Bright red rectal bleeding

B) Vomiting blood

C) Black tarry stool

D) Occult blood in stool

Answer: C) Black tarry stool

46. In Hypothyroidism, TSH is:

A) Low (↓)
B) Normal
C) High (↑)
D) Absent

Answer: C) High (↑)

47. In Hyperthyroidism, body weight:

A) Increases
B) Stays the same
C) Decreases
D) Fluctuates

Answer: C) Decreases

48. Which maneuver helps convert SVT (Supraventricular Tachycardia)?

A) Breathing deeply and fast
B) Valsalva Maneuver
C) Jumping jacks
D) Drinking hot water

Answer: B) Valsalva Maneuver

49. Penumbra in Ischemic Stroke refers to:

A) The core dead tissue
B) Salvageable tissue around the clot
C) The hemorrhagic area
D) Cerebrospinal fluid

Answer: B) Salvageable tissue around the clot

50. The first test to order for ANY new ICU patient is:

A) ABG
B) ECG
C) Random Blood Sugar
D) Complete Blood Count

Answer: C) Random Blood Sugar

🏆 Good Luck in Your Final Exam! بالتوفيق في الامتحان 🏆